

PROFESSIONAL SUMMARY

Gameplay Programmer and VR Developer with experience building interactive gameplay systems using Unity, C++, and C#. Skilled in VR interaction systems, procedural generation, gameplay mechanics, and AI behaviour systems. Passionate about designing technical systems that support immersive and narrative-driven player experiences.

TECHNICAL SKILLS

Game Engines

Unity, Unreal Engine

Programming Languages

C++, C#, Python

Gameplay & Systems

Gameplay Systems, VR Interaction Systems, Procedural Generation, Finite State Machines, AI Behaviour Systems

Tools & Technologies

Git, Unity XR Toolkit, NavMesh, Unity UI Toolkit, Jira, Confluence

3D & Design Tools

Blender, Unreal UMG, Figma

GAME DEVELOPMENT PROJECTS

A Day in My Life — VR Narrative Puzzle Game

Unity | C# | Blender | XR Toolkit | Oculus VR | 2025

- Developed a 3-scene VR puzzle experience in Unity (XR Interaction Toolkit).
- Implemented object grabbing, fragment rotation, teleport locomotion, and climbing mechanics in C#.
- Programmed two puzzle systems (leaf assembly, spiral alignment) using snapping and trigger detection.
- Engineered a Narrative Director system to trigger narration, subtitles, and scene progression.
- Optimized VR scene performance by replacing high-cost 3D grass assets with lightweight textures.
- Evaluated gameplay through peer playtesting (4.6/5 atmosphere, 4.2/5 immersion).

Treasure Sea-ker — 3D Underwater Platformer

Unity | C# | Unity UI Toolkit | GitHub | 2024 | Treasure Sea-ker WebGL Build- <https://jsrgolu.itch.io/treasure-sea-ker>

- Collaborated in a 4-person team to develop a 3-level underwater platformer with AI enemies and inventory systems in Unity.
- Implemented gameplay UI systems including inventory, minimap, and HUD using Unity UI Toolkit.
- Programmed player controls including movement, sprinting, jumping, and weapon switching in C#.
- Developed an inventory system supporting collectible, usable, and equipable items.
- Implemented enemy AI behaviour using a 5-state FSM (Idle, Patrol, Chase, Attack, Dead) with NavMesh.
- Added scalable difficulty modes adjusting enemy health, damage, and spawn counts.
- Improved gameplay through iterative playtesting (functionality score 7.31 → 8.72).

Procedural Forest Generator with Autonomous Agents

Unity | C# | 2025

- Built a procedural forest generator on a 124×70 grid using cellular automata.
- Implemented a three-stage generation pipeline (random fill, smoothing, region cleanup).
- Generated exploration paths using a drunken walk algorithm.
- Programmed 3 autonomous wandering agents with configurable speed and waypoint movement.
- Configured forest density (85%) and dynamic tree placement for varied map generation.

Dual-Control — Custom 2D Game Engine Project

C++ | Visual Studio Code | 2025

- Built a 2D action game from scratch in C++ without a game engine, implementing core gameplay systems.
- Developed systems for character movement, input handling, and collision detection.
- Implemented a dynamic input-switching system allowing real-time toggling between control schemes.
- Focused on memory management and low-level event handling, demonstrating understanding of engine fundamentals.

PROFESSIONAL EXPERIENCE

Technical Support Analyst

Tata Consultancy Services, Mumbai | Client: Westpac Banking Corporation

Jul 2022 – Aug 2024

- Resolved backend incidents for Westpac's CBF (Customer Business First) banking platform supporting enterprise banking operations.
- Monitored scheduled batch jobs responsible for financial data transfers across production systems.
- Diagnosed and recovered failed data pipelines using SQL scripts and backend troubleshooting.
- Investigated system errors and performance issues through log analysis and debugging.
- Documented incident recovery procedures and operational workflows to support faster troubleshooting and team knowledge sharing.
- Worked in Agile support environments using Jira and Confluence for incident tracking and documentation.

EDUCATION

Queen Mary University of London, Mile End, United Kingdom

(Sep 2024 – Sep 2025)

Msc Computer Games

University of Mumbai- Vishwaniketan's Institute of Technology, Khopoli, Raigad, India

(Aug 2019 – Jun 2022)

BE in Computer Science

(CGPA:7.86/10)

COMMUNITY & ACTIVITIES

- Participated in game jams on itch.io, collaborating on rapid game prototypes.
- Attended AI in Games conferences to explore emerging technologies in game development.
- Member of the Women in Games community.
- Completed training in animation, rigging, and rendering using Blender.
- Active in university arts community, with artwork featured on the college social media page.